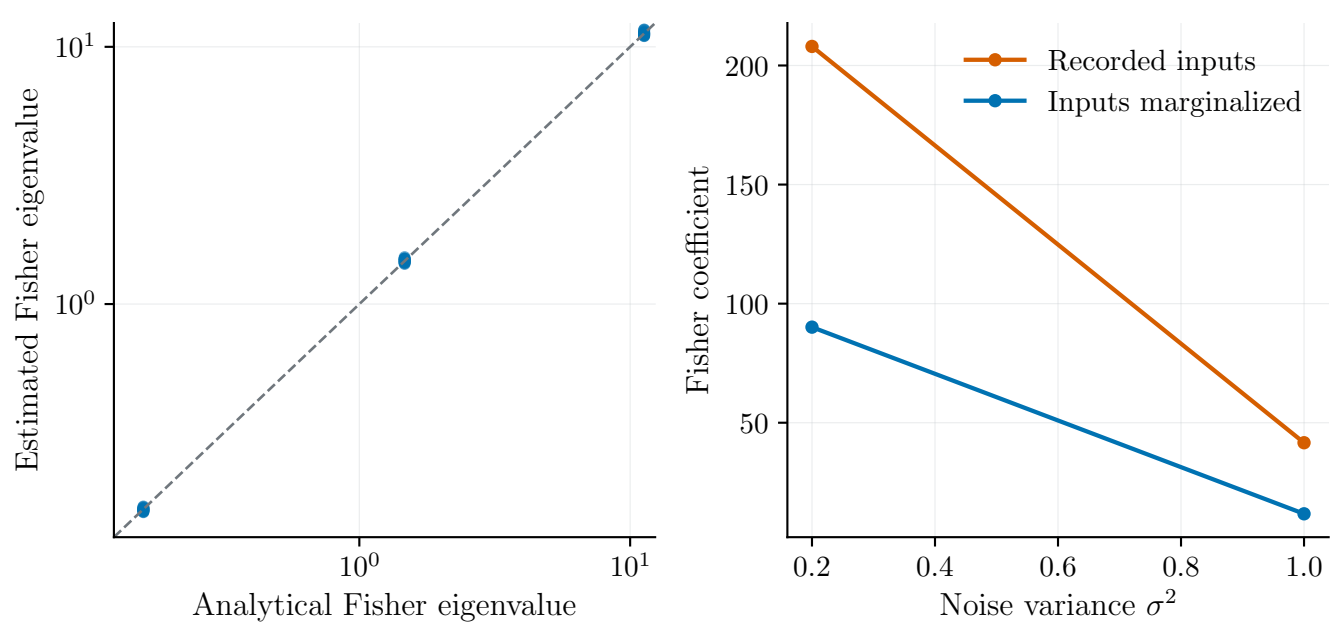
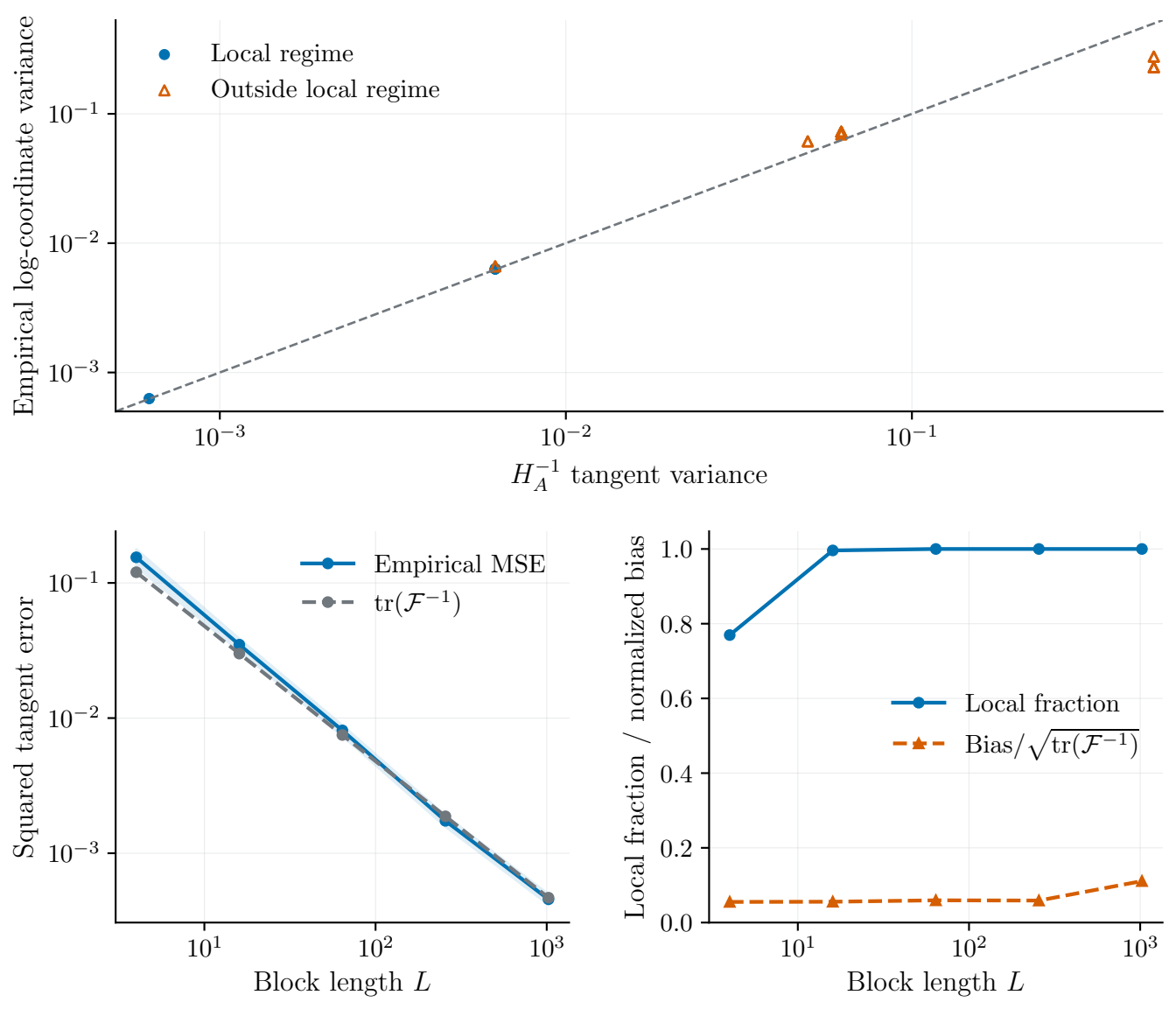
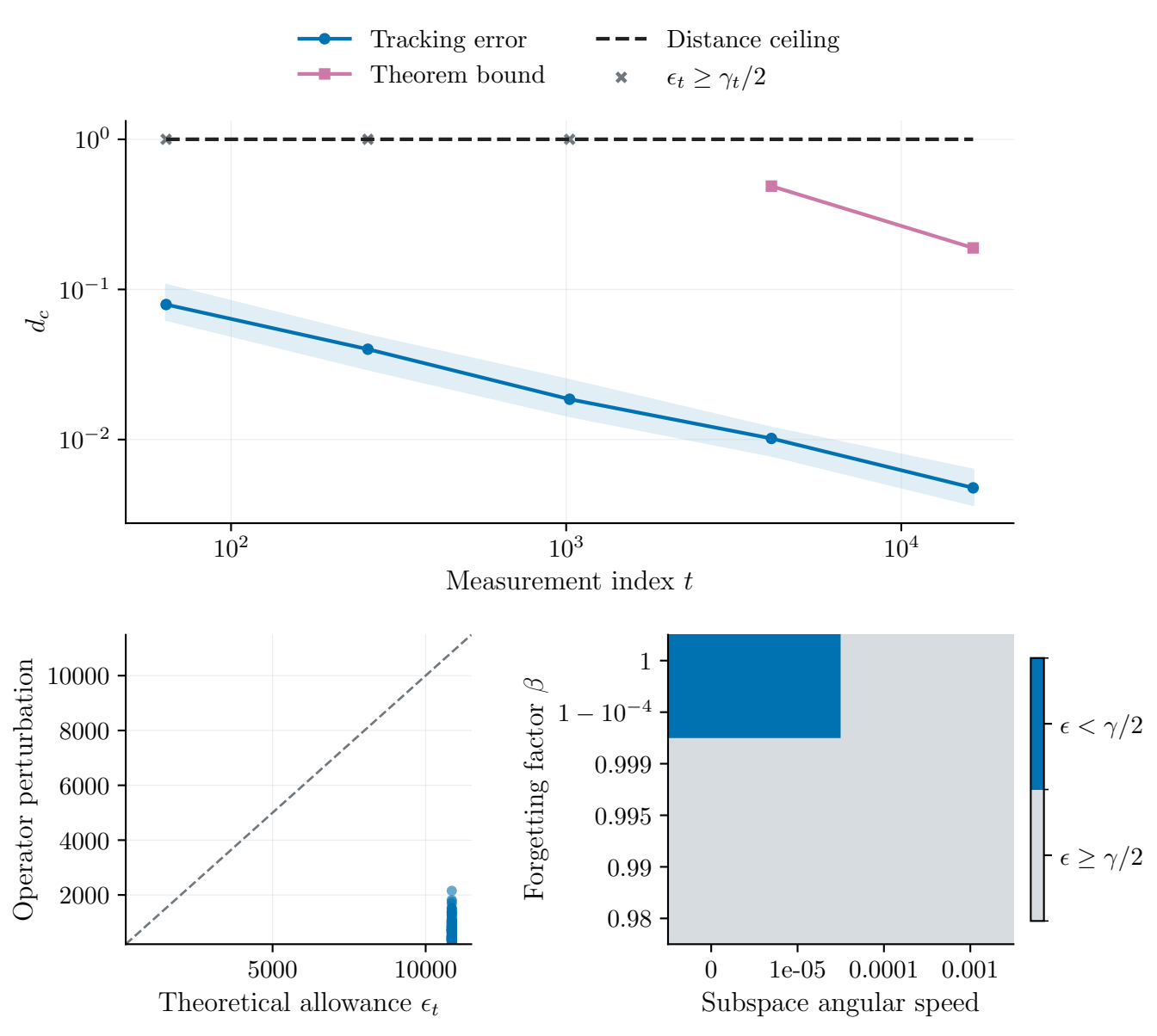
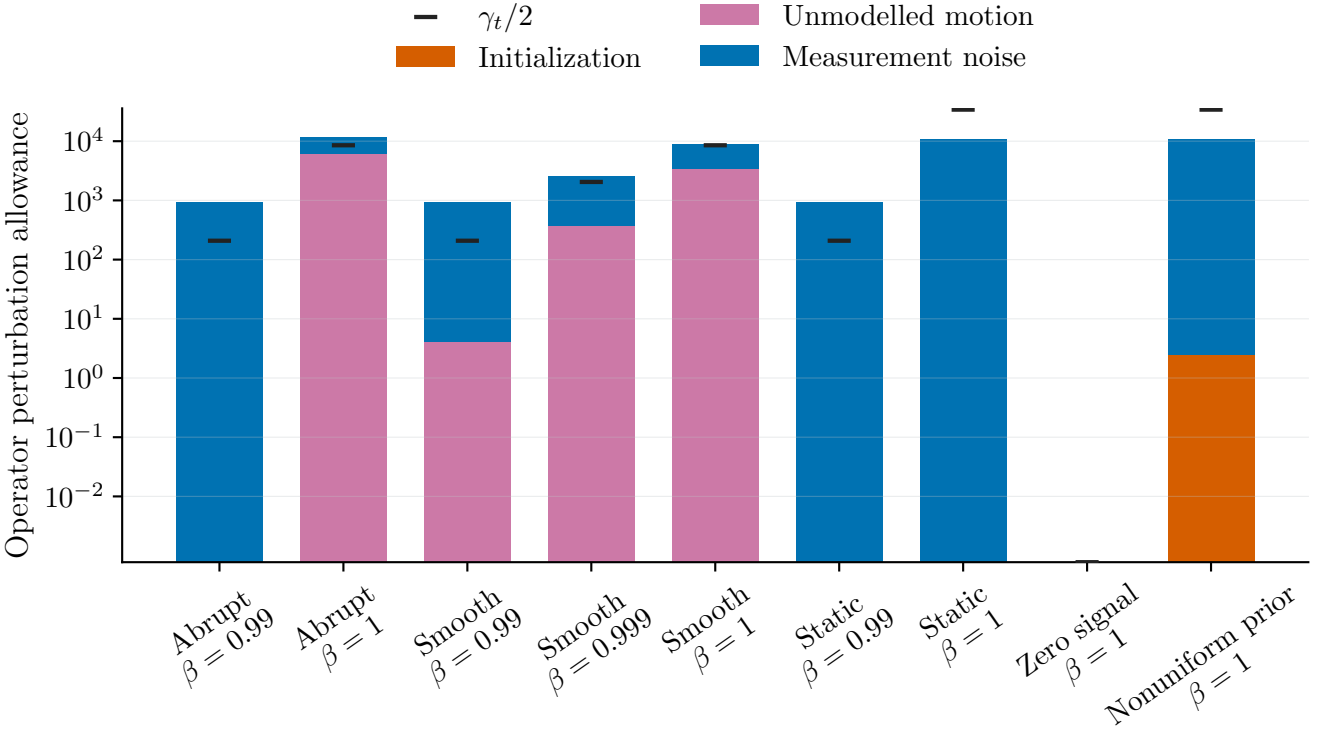
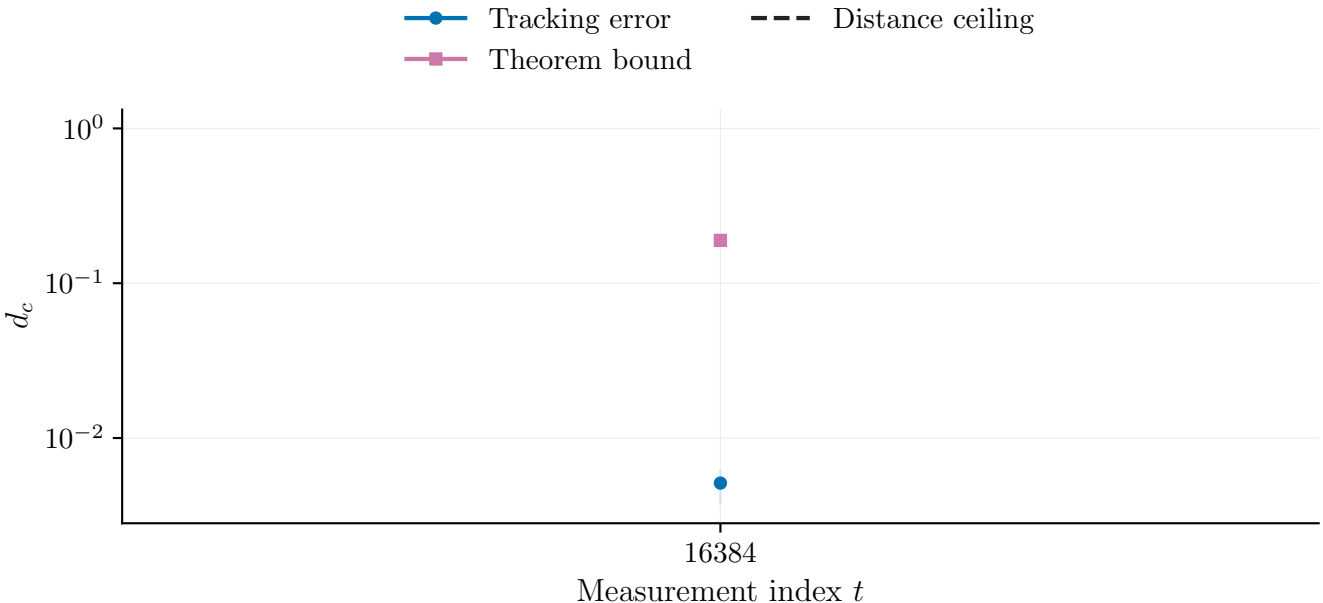
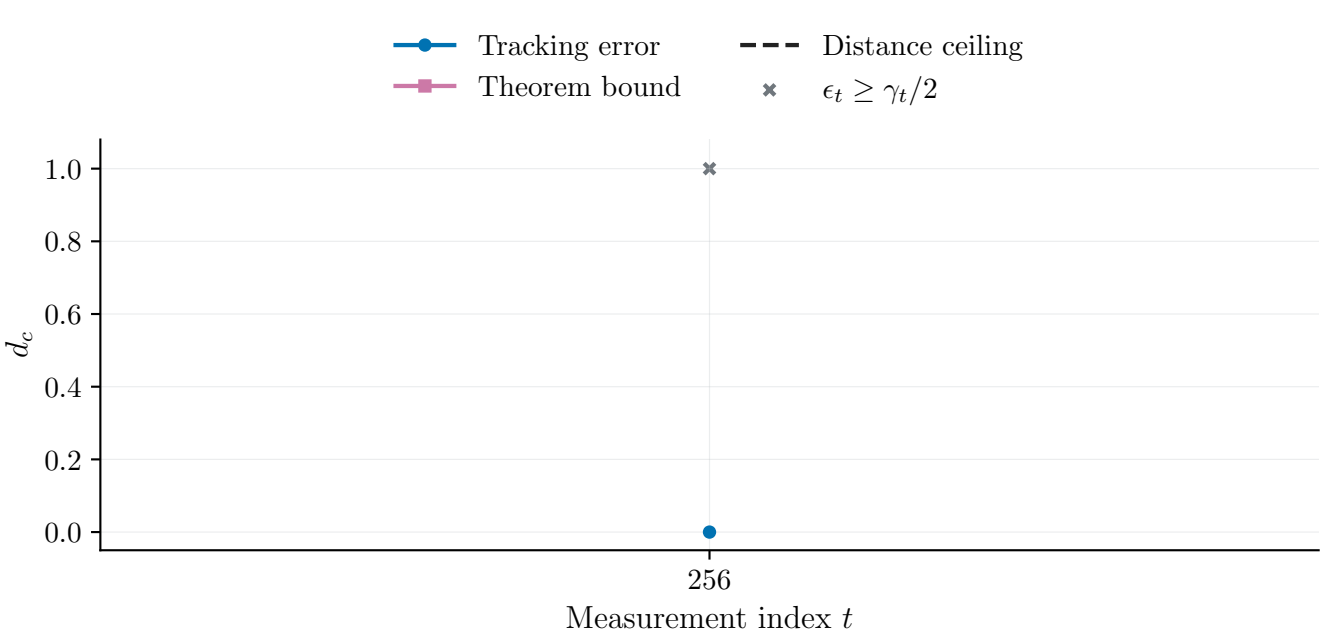

10^1
10^0
Estimated Fisher eigenvalue
10^0
10^1
Analytical Fisher eigenvalue
200
150
100
50
Fisher coefficient
Recorded inputs
Inputs marginalized
0.2
0.4
0.6
0.8
1.0
Noise variance σ^2


Local regime
Outside local regime
Empirical log-coordinate variance
10^{-1}
10^{-2}
10^{-3}
10^{-3}
10^{-2}
10^{-1}
H_A^{-1} tangent variance
Empirical MSE
$\mathrm{tr}(\mathcal{F}^{-1})$
Squared tangent error
10^{-1}
10^{-2}
10^{-3}
10^{1}
10^{2}
10^{3}
Block length L
Local fraction
Bias/$\sqrt{\mathrm{tr}(\mathcal{F}^{-1})}$
Local fraction / normalized bias
1.0
0.8
0.6
0.4
0.2
0.0
10^{1}
10^{2}
10^{3}
Block length L


Tracking error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
10^0
10^{-1}
10^{-2}
d_c
10^2
10^3
10^4
Measurement index t
Operator perturbation
10000
8000
6000
4000
2000
5000
10000
Theoretical allowance ϵ_t
Forgetting factor β
1
$1 - 10^{-4}$
0.999
0.995
0.99
0.98
0
1e-05
0.0001
0.001
Subspace angular speed
$\epsilon < \gamma/2$
$\epsilon \geq \gamma/2$


$\gamma_t/2$
Initialization
Unmodelled motion
Measurement noise
Operator perturbation allowance
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
Abrupt $\beta = 0.99$
Abrupt $\beta = 1$
Smooth $\beta = 0.99$
Smooth $\beta = 0.999$
Smooth $\beta = 1$
Static $\beta = 0.99$
Static $\beta = 1$
Zero signal $\beta = 1$
Nonuniform prior $\beta = 1$


Tracking error
Theorem bound
Distance ceiling
10^{0}
10^{-1}
10^{-2}
d_c
16384
Measurement index t


Tracking error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
1.0
0.8
0.6
0.4
0.2
0.0
d_c
256
Measurement index t